Tester Present

Specialist Automotive Solutions

TESTER ENGINEERING SUITE · V1.0.9-2026

POWER-USER ADDENDUM

DEVELOPER & ADVANCED DIAGNOSTICS

The two hidden menus: Developer (Ctrl+Shift+E) & Advanced Diagnostics (Ctrl+Shift+A).

"Two hidden menus for people who already know what the items do."

APPLIES TO Tester Engineering Suite v1.0.9-2026
COMPANION Keep alongside the standard User Manual
WEB tester.engineering



Right to Repair

BUILT BY A MECHANIC, FOR THE RIGHT TO REPAIR

READ THIS BEFORE YOU CLICK

TWO HIDDEN MENUS

The suite reserves two menus for power users. Both are invisible by default, both unlock with a keyboard chord, and both are intended for technicians who already understand what the items underneath them do. Some items are destructive — read the relevant section before clicking.

MENU	CHORD	COLOUR	IT INSPECTS…
Developer	Ctrl + Shift + E	Purple	the suite itself — licence cache, transport state, storage, MCP, telemetry.
Advanced Diagnostics	Ctrl + Shift + A	Dark blue	the vehicle — bus health, transport debugging, structured fault-finding.

NOTE

Neither menu persists. The chord shows it; the same chord hides it; relaunching the suite hides it again. This keeps a regular operator from stumbling into either one.

01 DEVELOPER MENU

Ctrl + Shift + E — a purple 'Developer' item appears on the menu strip.

Where the Advanced menu inspects the vehicle, the Developer menu inspects the suite: session state, transport, licence cache, storage, the automation server and build info.

Secret vehicle workspaces

ITEM	WHAT IT DOES
Allow Secret Coyote	Reveals the Coyote PCM workspace (diagnostics / configuration / flash for the supported Coyote powertrain module) without needing a matching VIN.
Allow Secret Ranger	Reveals the Ranger/Everest diesel ECM workspace without needing a matching VIN. For bench work and preparation.

Session & State

ITEM	WHAT IT DOES
Show Current Session State…	Dumps VIN, TX/RX IDs, transport selection, J2534 firmware, frame counters and DTC counters into a copyable report dialog.
Reset CAN-Known-Dead Flag	Clears the sticky 'CAN bus is unresponsive' decision so the next Live / Signal Analyser start re-probes from scratch.
Reset Current VIN	Wipes the in-memory VIN; the next read overwrites it.
Show Active Transport	Reports user preference, currently active transport, K-Line state and the CAN-known-dead flag.

Connection

ITEM	WHAT IT DOES
List J2534 Devices	Full table of every J2534 PassThru DLL the device finder discovered, plus serial-port ELM327 entries.
Probe ECU Address…	Send an arbitrary service-ID byte to a chosen TX/RX pair and watch the response — a cheap reachability test before opening a full session.
Test Commands (PassThru / AT)…	The raw J2534 / ELM327 console: manual PassThru calls, AT command stream, and the request/response log.
CAN Bus Health Check (J2284-3)…	Single-shot health probe — termination check, idle-frame distribution and fault classification per SAE J2284-3 §5.6 Table 1.
Check CAN Log for TES Fingerprint…	Scan a .log / .asc capture for the integrity fingerprint the suite writes during fingerprinted sessions. Used in forensics.
Fingerprinting active (toggle)	Master switch for session traffic fingerprinting (ISO-TP padding + cadence + probe on connect). On by default.
Idle CAN-Log Scanner (toggle)	Background filesystem watcher that looks for the fingerprint in third-party CAN logs. Requires EULA consent + the idle-scan consent flag before it will start.

ITEM	WHAT IT DOES
Idle Scanner Status…	Shows scanner state, last directory walked, file count and any matches found this session.

License & Killswitch

ITEM	WHAT IT DOES
Show Cached RemoteConfig…	Reveals the decrypted contents of the offline licence cache (server URL, API key, signing key, expiry) for this machine.
Re-fetch RemoteConfig (online)	Forces a fresh online fetch from the licence server and rewrites the local cache.
Clear License Cache	Wipes the cached config. The next launch starts cold and must reach the licence server.

NOTE

These items expose only your own machine's licence material for support/debugging. They carry no vehicle data — the licence path and vehicle data are separated by design.

Storage

ITEM	WHAT IT DOES
Open Vehicle History Folder	Opens ~/Documents/Tester Present History — birth certificates, flash certificates, receipts.
Open AppData / TesterPresent	Opens %LOCALAPPDATA%/TesterPresent — preferences, licence cache, scratch files.
Open Build Output Folder	Opens the directory containing the running EXE.

UI Diagnostics

ITEM	WHAT IT DOES
List Tab Pages	Every tab in the main window with its index and visibility state — useful for confirming VIN-gated tabs show/hide correctly.
Test All Toast Types	Fires Info / Success / Warning / Error toasts in sequence so styling can be QA'd.
Force GC + Show Memory Stats	Forces a garbage collection, then reports working set, managed heap and GC generations — for chasing leaks in long sessions.
Force-show Tesla tab	Forces the Tesla workspace visible without a Tesla VIN, for bench testing that UI.
Force-show Falcon tab	Forces the Falcon workspace visible without a matching VIN.
Heavy-vehicle workspaces ►	Auto (follow the selected adapter, default) · Always show (bench, no adapter needed) · Always hide (RP1210 + heavy interfaces off).

Build Info

ITEM	WHAT IT DOES
Show Version & Build Info	Assembly version, build timestamp, target framework, runtime, OS and machine name.
Copy Diagnostic Bundle to Clipboard	A single text blob of every diagnostic dump above — paste it straight into a support ticket.

Power & Battery

ITEM	WHAT IT DOES
Show Power Status…	AC power state, battery percentage, lid state and the active execution-state scope

ITEM	WHAT IT DOES
	(the suite holds the PC awake during long operations).
Open Windows Power Options	Shortcut to the Windows power control panel.
USB Suspend Field Notes…	Reference text on USB selective-suspend behaviour and J2534 PassThru reliability.

Police Scanning

ITEM	WHAT IT DOES
Background 00:25:DF Scanner	Opens the background Bluetooth-LE watch that alerts on a specific device-address signature in range.
Police Scanner active (toggle)	Runs that BLE watch in the background. Requires Windows Bluetooth enabled and the Location service running.

NOTE

A situational-awareness tool, not a diagnostic one. It reads only Bluetooth advertising already broadcast in the open air; it stores nothing about vehicles or people.

MCP Server — local automation bridge

The suite can expose a local Model-Context-Protocol server so an AI assistant on the same machine can drive the documented diagnostic tools (connect, read VIN/DTCs/PIDs, run dry-runs, and the rest) through a controlled interface. It is off unless you turn it on, token-authenticated, and gated by a per-session allow-list.

ITEM	WHAT IT DOES
MCP server running (toggle)	Starts / stops the local MCP server.
Show Connection Info…	Displays the local endpoint the assistant should connect to.
Copy Auth Token to Clipboard	Copies the current auth token for pasting into the assistant's config.
Show Claude Desktop Config Snippet…	Prints a ready-to-paste MCP client configuration block.
Rotate Auth Token	Issues a fresh token and invalidates the old one.
Restart MCP Server	Restarts the server (e.g. after rotating the token).
Open MCP Log Folder	Opens the server's log directory.
Show MCP Tool List…	Lists the diagnostic tools currently exposed to a connected assistant.
Clear Session Allow-list	Revokes every action the assistant was granted this session; it must ask again.

WARNING **The MCP bridge can act on a vehicle**

The exposed tools include live-vehicle actions. Only enable the server on a machine you control, share the token with nothing you do not trust, and clear the allow-list when you are done. Treat it as handing someone a second keyboard.

02 ADVANCED DIAGNOSTICS

Ctrl + Shift + A — a dark-blue 'Advanced Diagnostics' item appears on the menu strip.

Thirty-plus engineering tools grouped into nine sub-menus. Read-only reference and analysis tools are safe to open anywhere; tools that transmit need a live connection and are noted as such. Many need a J2534 device (not ELM327) and an active UDS or OBD session.

Bus & Transport

ITEM	WHAT IT DOES
CAN Bus Health Check (J2284-3)…	Single-shot termination / dominant-recessive / error-counter classification (Type A/B/C). The right first move when U-codes are present.
AT Commands / PassThru Console…	Raw transport console — switches between J2534 PassThru calls and ELM327 AT commands. Verify the device answers before suspecting the vehicle.
CANbus Diagnosis (live)…	Continuous frame stream: live counts of frames, FPS, unique IDs, estimated bus load and error counter, with a top-20 dominant-ID table and hints. Catches slow-burn faults the single-shot test misses.
ISO-TP Frame Disassembler…	Breaks multi-frame ISO-TP traffic into its single/first/consecutive/flow-control frames for inspection.
CAN Bit-Timing Analyzer…	Sample-point and bit-timing analysis for a bus that only works sometimes.
Ground-Loop Detector (multimeter-guided)…	Guided procedure to find ground-offset / ground-loop faults with a multimeter.
ELM327 Comparator (two devices)…	Runs the same requests through two adapters side by side to expose a flaky or non-conforming clone.

Fault Diagnosis

ITEM	WHAT IT DOES
Manual DTC Lookup…	Type or paste a code (P0420, U0100, B1601, C0035…) for its J2012 family description, severity hint and triage walkthrough. Licensed OEM codes decode to manufacturer text; others return the generic family.
Fault-Finding Walkthrough…	A wizard for 'no idea where to start' cases: pick a symptom and read a structured decision tree — what to ask the customer, the cheapest first reading, and the trap most techs fall into.
Symptom → DTC Reverse Search…	Work backwards from a symptom to the codes and systems most likely responsible.
DTC Snapshot / Diff (Pre/Post Repair)…	Capture codes before a repair and again after, then compare. Needs a live session.
DTC Timeline Visualiser (Gantt)…	Plots when codes set/cleared across a session as a timeline.
Permanent DTC Browser (Mode $0A)…	Reads permanent codes that a clear cannot erase until the monitor re-runs.
Drive Cycle / Readiness Trainer…	Guides the drive cycle needed to complete readiness monitors.

Live Data

ITEM	WHAT IT DOES
PID Logger / Chart Recorder…	Multi-channel PID logging with a chart recorder view. Needs a live OBD session.
Long-Term Fuel Trim Trender…	Trends LTFT over time to expose mixture drift.
O2 Sensor Cross-Counts Analyzer…	Counts O2 sensor switch events to judge sensor health.
Misfire Counter Live…	Per-cylinder misfire counts as they happen.
Live Combustion Map (RPM × Load × Trim)…	Builds a live map of fuel trim against RPM and load.
Live RAM Probe (KWP 21 LL / UDS 22 DD)…	Reads live memory/DID values continuously. Needs a live UDS/KWP session.

Engine Diagnostics

ITEM	WHAT IT DOES
Relative Compression from CKP…	Ranks cylinder contribution from crank-sensor ripple — a compression picture without a gauge.
Cylinder Balance (UDS RoutineControl)…	Runs a cylinder-balance routine and reports per-cylinder contribution.
Adaptation-Reset Quick Actions…	Common adaptation/relearn resets in one place. Confirms before it writes.

Identification

ITEM	WHAT IT DOES
Offline VIN Decoder…	Decode any VIN into WMI / VDS / VIS and vehicle attributes without a connection.

Network Forensics

ITEM	WHAT IT DOES
Diagnostic Audit Trail…	Reviews the audit trail of diagnostic actions taken this session.
Module Topology Animation…	Animated view of the module network and which modules sit on which bus.

Reverse Engineering

ITEM	WHAT IT DOES
Seed-Key Sandbox…	Work out and verify a security-access key against a known seed offline, before spending a lockout-counted attempt on a live module.
Ghost ECU — bench CAN responder…	Turn the interface into a bench ECU that answers diagnostic requests — for testing tools and harnesses without a real module.

Reference

ITEM	WHAT IT DOES
NRC Decoder…	Decode a UDS negative-response code (7F …) into plain English.
UDS Service & DID Dictionary…	Searchable reference of UDS services and common data identifiers.
Vendor Address Map…	Per-OEM CAN address reference for common modules.

ITEM	WHAT IT DOES
J1962 Pinout Visualiser…	The OBD-II connector pinout, annotated per protocol.
ISO 15765 Frame Cheat Sheet…	Quick reference for ISO-TP framing.

Reporting

ITEM	WHAT IT DOES
Customer-Facing Diagnostic Report…	Builds a clean, customer-ready report from the current session. Needs a live session.

03 OPERATING NOTES

How to run these menus without hurting anything.

Both menus are transient

Show either menu with its chord; hide it with the same chord. Nothing persists — relaunching the suite hides both again, so a regular operator will not stumble into them.

Read-only vs transmit

The reference and analysis tools (manual DTC lookup, fault-finding walkthrough, NRC decoder, dictionaries, offline VIN decoder, pinout, cheat sheets) are safe to open anywhere. The console, probe, bench-responder and any tool marked 'live' can put traffic on the bus — treat those like a torque wrench, not a hammer.

Live-connection tools

Many Advanced Diagnostics tools need a live connection and a session: a real J2534 device (not ELM327), and often an active UDS or OBD session. If a tool opens empty or greyed, check the status bar — usually the session or the transport is the reason.

Licensed vs generic DTC data

Manual DTC Lookup and the OEM decoders use bundled databases unlocked by your active licence. Without an active licence the lookup falls back to generic SAE J2012 family decoding — you still get a family and severity, just not the manufacturer-specific text.

The safety rules still apply here

> **WARNING** **These menus do not lift the guard rails**
>
> The restraint/airbag hard-block, the flash safety supervision, and the confirm-before-irreversible prompts all still apply to anything you launch from these menus. The Advanced tools give you more reach, not fewer rules. Bench-only workspaces exposed by the Developer menu (Coyote, Ranger, Tesla, Falcon, heavy) are for preparation and testing; validate on the bench before you rely on them against a vehicle.

Where to find more: the standard User Manual covers the everyday workspaces; tester.engineering has the current capability list. Keep this addendum with the manual.